

GARCH and stochastic volatility models

Christian Francq

Chapter 1: GARCH processes

AutoRegressive Conditional Heteroskedasticity, or AutoRegressive Conditionally Heteroskedastic Models of Engle* (1982):

"Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of U.K. Inflation," *Econometrica*, 50, 987–1008.

This model shows that conditional heteroskedasticity is compatible with marginal homoscedasticity (and even stationarity).

Generalized ARCH of Bollerslev (1986):

"Generalized Autoregressive Conditional Heteroskedasticity," *Journal of Econometrics*, 31, 309–328.

*Nobel 2003

Main application: modelling the **volatility** (variability) of asset returns model, used for **risk management** and financial regulation (conditional Value-at-Risk evaluation) and asset pricing/**portfolio selection** (building portfolio with optimal mean-variance), dynamic **beta hedging** (reducing the systematic risk by building a portfolio of assets with offsetting betas).

References:

- Brockwell and Davis (1991) *Time Series: Theory and Methods* Springer
- Francq and Zakoïan (2010, 2019) *GARCH Models: Structure, Statistical Inference and Financial Applications* Wiley
- Tsay (2010) *Analysis of Financial Time Series* Wiley
- Pamplemousse + <http://christian.francq140.free.fr/>

Example of application

Let a time series (e.g. a series of financial returns)

$$\epsilon_t = m_t + \sigma_t \eta_t$$

with conditional mean $m_t = E(\epsilon_t | \mathcal{F}_{t-1})$ and

$$\text{conditional variance—volatility—}\sigma_t^2 = \text{var}(\epsilon_t | \mathcal{F}_{t-1}),$$

where $\mathcal{F}_t$ is the information available at time t . Assume η_t iid (0,1), η_t and $\mathcal{F}_{t-1}$ independent. Risk measures (such as VaR or Expected Shortfall) generally satisfy

$$r(\epsilon_t | \mathcal{F}_{t-1}) = m_t + \sigma_t r(\eta_1).$$

For financial returns, m_t is often constant (and very close to 0, in accordance with the efficient-market hypothesis of Fama (1970)), but σ_t is highly time-varying (estimated by a [GARCH-type model](#)).

Another example of application

Let a stationary multivariate time series $(y_t, \mathbf{x}_t')'$.

Problem: Given $\mathcal{F}_{t-1}$, regress y_t on $\mathbf{x}_t$ (e.g. for beta-hedging with a CAPM model).

Solution: $y_t - E(y_t | \mathcal{F}_{t-1}) = \beta'_{y\mathbf{x},t} \{\mathbf{x}_t - E(\mathbf{x}_t | \mathcal{F}_{t-1})\} + \eta_t$, with the **dynamic conditional beta** $\beta_{y\mathbf{x},t} = \Sigma_{\mathbf{xx},t}^{-1} \Sigma_{\mathbf{xy},t}$.

For implementation: A multivariate GARCH-type model for the conditional variance $\begin{pmatrix} \Sigma_{\mathbf{xx},t} & \Sigma_{\mathbf{xy},t} \\ \Sigma_{y\mathbf{x},t} & \Sigma_{yy,t} \end{pmatrix}$ of $\begin{pmatrix} \mathbf{x}_t - E(\mathbf{x}_t | \mathcal{F}_{t-1}) \\ y_t - E(y_t | \mathcal{F}_{t-1}) \end{pmatrix}$ is needed.

Few alternatives to discrete time volatility models

- ① Rolling windows (silly but widely used approach)
- ② Filters such as Exponential Moving Average (EMA in Riskmetrics): $\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) r_t^2$ with $\lambda = 0.94$
- ③ The VIX based on implied volatilities of S&P 500 option prices
- ④ Diffusion processes (continuous-time Markovian processes)
- ⑤ Realized volatilities of intraday returns: if the continuous-time log-return at time $u \in [0, 1]$ of day t is

$$r_t(u) := \ln p_t(u) / p_t(0) = \sigma_t W_t(u)$$

where $W_t(\cdot)$ is a standard Brownian motion, then

$$\sigma_t^2 = \lim_{h \rightarrow 0} \sum_{k=1}^{[1/h]} \{r_t(kh) - r_t(kh - h)\}^2 \text{ in } L^2$$

- ⑥ Machine-learning approaches, ...

Outline of the course

- ① GARCH processes
- ② Asymmetric GARCH
- ③ Statistics of GARCH models
- ④ Multivariate GARCH models
- ⑤ Stochastic Volatility Models

Evaluation rules and objectives

- Standard exam of two hours, without document;
- A (too large) number of corrected past exams is available;
- The objectives are to:
 - demonstrate that GARCH-type models mimic several stylized facts of financial returns;
 - apply basic concepts of time series (stationarity, invertibility, inference, prediction, ...) within the GARCH framework;
 - acquire a certain level of technicality[†];
 - see how such volatility models can be employed in practice;
 - convince it is easy and quite efficient in practice to use GARCH-type models.

[†]the added value of ENSAE students, compared to other schools

Outline of the chapter

- 1 Financial series and GARCH models
 - Stylized facts on financial series
 - Models for conditional heteroskedasticity
 - GARCH processes
- 2 Stationarity conditions
 - Weak ARMA representation
 - Stationarity of the GARCH(1,1)
 - Stationarity of the GARCH(p, q)
- 3 Other properties
 - Marginal moments
 - Autocorrelations of squared returns
 - Forecasts

ARMA and GARCH models are not competitors but complementary

- Linear models (Wold, 1938) and ARMA models (Box and Jenkins, 1970)
 - Objective of ARMA models: modelling the conditional mean.
- ARCH models (Engle, 1982) and GARCH (Bollerslev, 1986)
 - Objective of GARCH models: modelling the conditional variance.

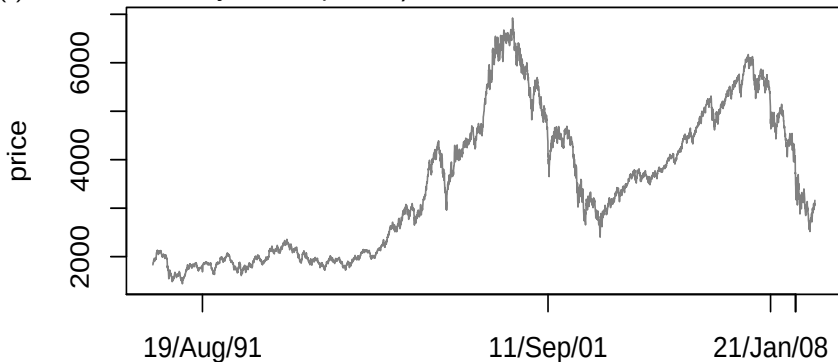
Main characteristics of the financial series :

- various types of series (stock prices, commodity prices, interest rates, exchange rates, ...), with different frequencies (second, minute, hour, day, etc), often with a huge number of observations.
- existence of "stylized facts", an econometric term for a set of empirical characteristics shared by most of the financial series (already listed by Mandelbrot in 1963).

Stylized facts (Mandelbrot (1963))

A large number of financial series exhibit similar properties

(i) *Non stationarity of the prices p_t .*



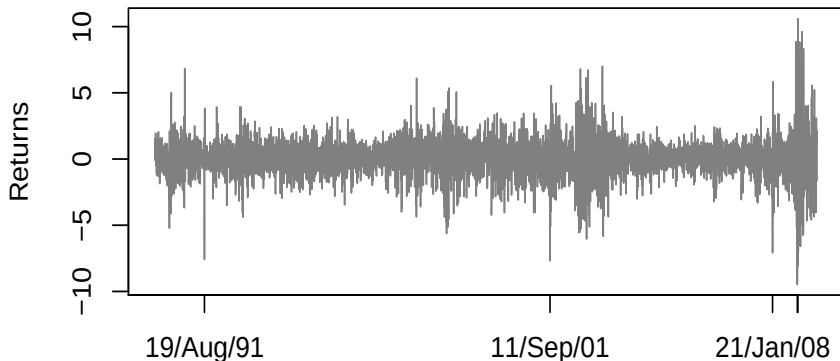
CAC 40, from 01/03/1992 to 30/04/2009

► More recent data

► SP 500

(ii) *Possible stationarity of the returns.*

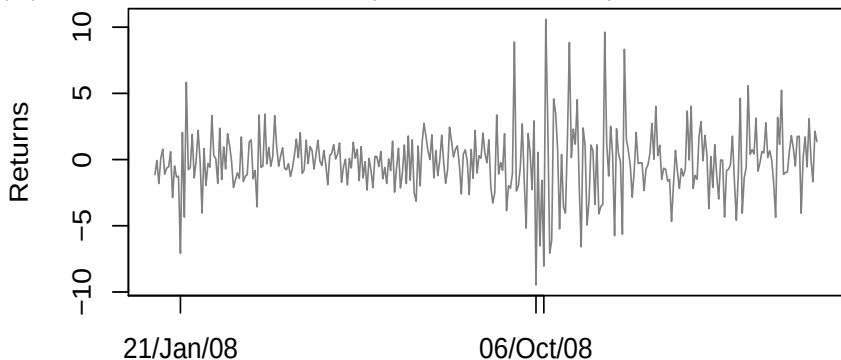
$$\epsilon_t = \log(p_t/p_{t-1}) \approx \frac{p_t - p_{t-1}}{p_{t-1}}$$



► SP 500

CAC 40 returns, from March 2, 1990 to February 20, 2009

(iii) *Cluster of extreme values (volatility clustering).*



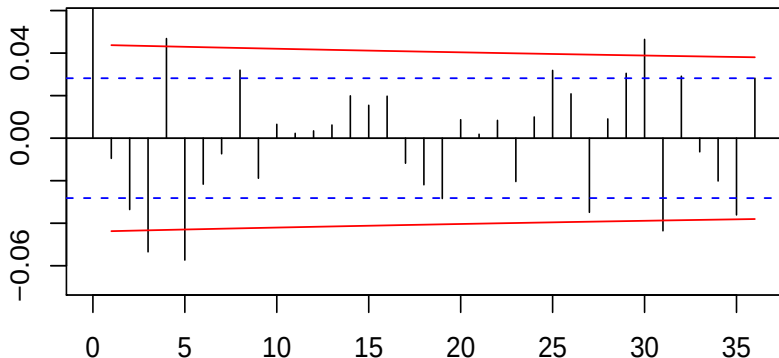
CAC 40 returns, from January 2, 2008 to April 30, 2009

► zoom SP500

Stylized Facts

Dependence without correlation (warning: the dotted lines $\pm 1.96/\sqrt{n}$ can be misleading)

(iv) *Uncorrelatedness of the returns.*

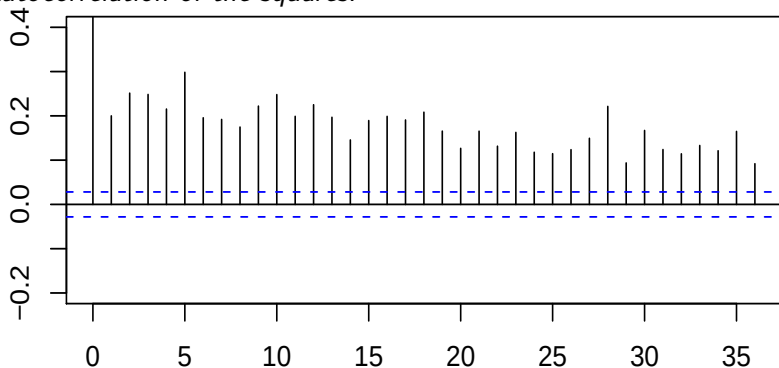


► SP 500

► Other indices

Empirical autocorrelations of the CAC returns

(v) *Autocorrelation of the squares.*



Autocorrelations of the squares of the CAC returns

► SP 500

Portmanteau tests

Autocorrelation Check for White Noise (on returns)

To Lag	Chi- Square	DF	Pr > Khi2	----- Autocorrelations -----					
6	11.51	6	0.0737	0.030	0.005	-0.032	0.028	-0.046	-0.001
12	16.99	12	0.1499	-0.018	-0.014	0.034	0.016	0.017	0.010
18	21.22	18	0.2685	-0.005	0.025	-0.031	-0.009	-0.003	0.006
24	27.20	24	0.2954	-0.023	0.003	-0.010	0.030	-0.027	-0.015

Autocorrelation Check for White Noise (on squared returns)

To Lag	Chi- Square	DF	Pr > Khi2	----- Autocorrelations -----					
6	165.90	6	<.0001	0.129	0.127	0.117	0.084	0.101	0.074
12	222.93	12	<.0001	0.051	0.060	0.070	0.092	0.058	0.030
18	238.11	18	<.0001	0.053	0.036	0.020	0.041	0.002	0.013
24	240.04	24	<.0001	0.006	0.024	0.013	0.003	0.001	-0.002

Portmanteau tests for the white noise assumption on the CAC 40 series (top array) and on the squared CAC returns (bottom array)

Box-Pierce Q-statistics: $Q_{BP} = n \sum_{h=1}^H \hat{\rho}^2(h) \sim \chi_H^2$ as $n \rightarrow \infty$, under the strong white noise assumption

Other stylized facts

(vi) *Fat-tailed distributions for the returns.*

▸ Leptokurticity

(vii) *Asymmetries.* A drop of the prices entails more volatility than an increase of the same magnitude.

▸ Leverage effects

(viii) *Seasonality.*

Models with random conditional variance

ARMA models are useless for return series because

$$\hat{\rho}(h) \simeq 0 \quad \forall h \neq 0. \quad \text{▶ ARMA models}$$

All the return models are of the form

$$\epsilon_t = \sigma_t \eta_t,$$

where

- (η_t) iid $(0, 1)$;
- the process $\sigma_t > 0$ is called volatility;
- η_t is independent of σ_t and of $\{\epsilon_u, u < t\}$.

Properties implied by the multiplicative form

$$\epsilon_t = \sigma_t \eta_t, \quad (\eta_t) \text{ iid } (0, 1), \quad \eta_t \text{ independent of } \sigma_t, \{\epsilon_u, u < t\}$$

Remarks:

- when $\text{Var}(\eta_1) < \infty$, it is not restrictive to assume $\text{Var}(\eta_1) = 1$.
- when these moments exist, we have

$$E\epsilon_t = 0$$

and

$$\gamma_\epsilon(h) = E(\eta_t \sigma_t \epsilon_{t-h}) = 0;$$

- if (ϵ_t) is stationary, this is a **white noise**;

Properties implied by the multiplicative form (continued)

$$\epsilon_t = \sigma_t \eta_t, \quad (\eta_t) \text{ iid } (0, 1), \quad \eta_t \perp \sigma_t \text{ and } \epsilon_u, u < t$$

Remarks:

- when $E\epsilon_t^4 < \infty$, the distribution of ϵ_t is **leptokurtic** because

$$\begin{aligned} \kappa_\epsilon &:= \frac{E\epsilon_t^4}{\{E(\epsilon_t^2)\}^2} = \kappa_\eta E \frac{\sigma_t^4}{\{E(\sigma_t^2)\}^2} \\ &= \kappa_\eta \left\{ 1 + \frac{\text{Var}\sigma_t^2}{\{E\sigma_t^2\}^2} \right\} > \kappa_\eta. \end{aligned}$$

Two main classes of volatility models

$$\epsilon_t = \sigma_t \eta_t$$

Type ARCH: $\sigma_t \in \sigma(\epsilon_u, u < t)$, for instance $\sigma_t = \omega + \alpha|\epsilon_{t-1}|$. Then

$$\sigma_t^2 = \text{Var}(\epsilon_t | \epsilon_u, u < t)$$

is the conditional variance.

Type stochastic volatility: The process σ_t has its own dynamics, for instance

$$\log \sigma_t = \omega + \alpha \log \sigma_{t-1} + \nu_t$$

where (ν_t) is a strong white noise independent of (η_t) .

GARCH(p, q)

(η_t) iid sequence, $E\eta_t = 0$, $E\eta_t^2 = 1$.

Definition

(ϵ_t) is a GARCH(p, q) process if it satisfies

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2, \quad \forall t \in \mathbb{Z} \end{cases}$$

$\omega > 0$, $\alpha_i \geq 0$ ($i = 1, \dots, q$) , $\beta_j \geq 0$ ($j = 1, \dots, p$).

Remarks

- If a **causal** solution[‡] exists

$$\begin{cases} E(\epsilon_t | \epsilon_u, u < t) = 0, & t \in \mathbb{Z}, \\ \sigma_t^2 = \text{Var}(\epsilon_t | \epsilon_u, u < t) = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2. \end{cases}$$

- To ensure $\sigma_t^2 > 0$, one can impose $\omega > 0$, $\alpha_i, \beta_j \geq 0$.
- If the series (ϵ_t) is stationary, this is a semi-strong white noise.

► Strict and second-order stationarity

► White noises

- The optimal prediction is 0 and σ_t^2 is the conditional variance of the prediction error. Its non constant form, *i.e.* the conditional heteroskedasticity, allows for "volatility clustering".

► Linear and nonlinear optimal predictions

► Innovations

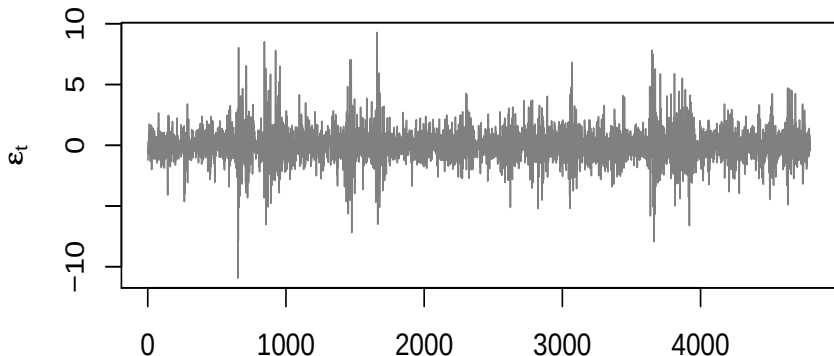
[‡]that is a solution such that σ_t be $\{\epsilon_u, u < t\}$ -measurable

Remarks (continued)

$$\begin{cases} E(\epsilon_t | \epsilon_u, u < t) = 0, & t \in \mathbb{Z}, \\ \sigma_t^2 = \text{Var}(\epsilon_t | \epsilon_u, u < t) = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2. \end{cases}$$

- With the GARCH(1,1) $\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, β is a **persistence** parameter, α is a **reaction** parameter to the last return, and ω is a **scale parameter** which has no impact on the relative shape of the volatility path. For most series of indices, α is close to 0.05 and β is close to 0.9. If a large squared return ϵ_t^2 appears, the next volatility σ_{t+1}^2 will increase and the effect will be highly persistent because β is close to 1.
- How to construct such GARCH processes ? are they stationary ? in particular, are conditional heteroskedasticity and marginal homoskedasticity compatible ?

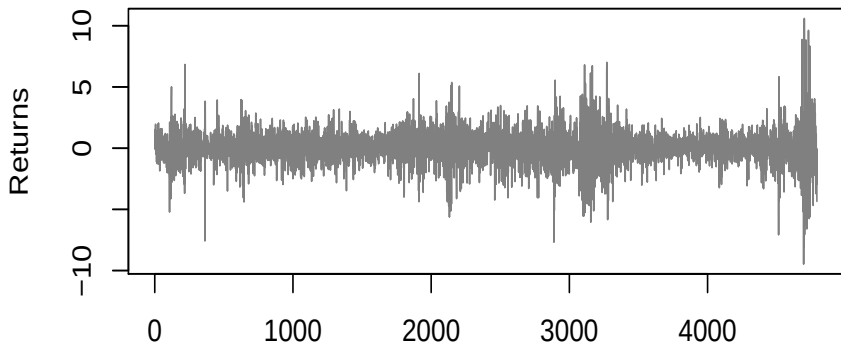
A simulated GARCH(1,1) trajectory



$$t = 1, \dots, n = 4791$$

$$\epsilon_t = \sigma_t \eta_t, \eta_t \text{ iid } St_5, \sigma_t^2 = 0.033 + 0.090\epsilon_{t-1}^2 + 0.893\sigma_{t-1}^2,$$

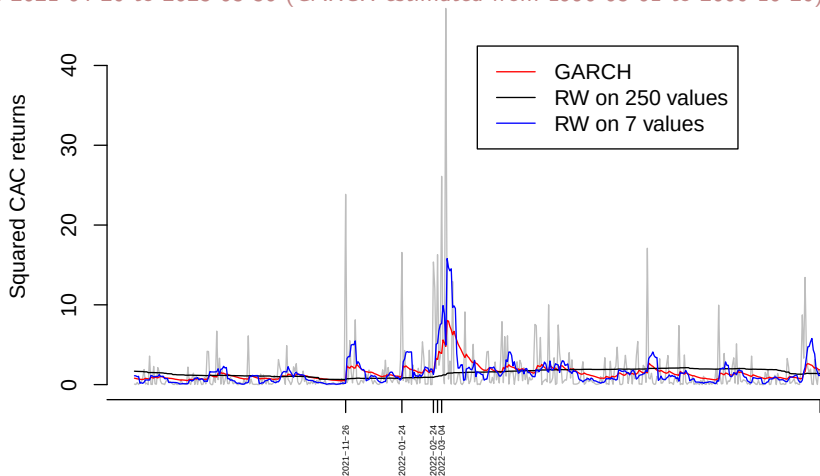
... which presents some similarities with return series



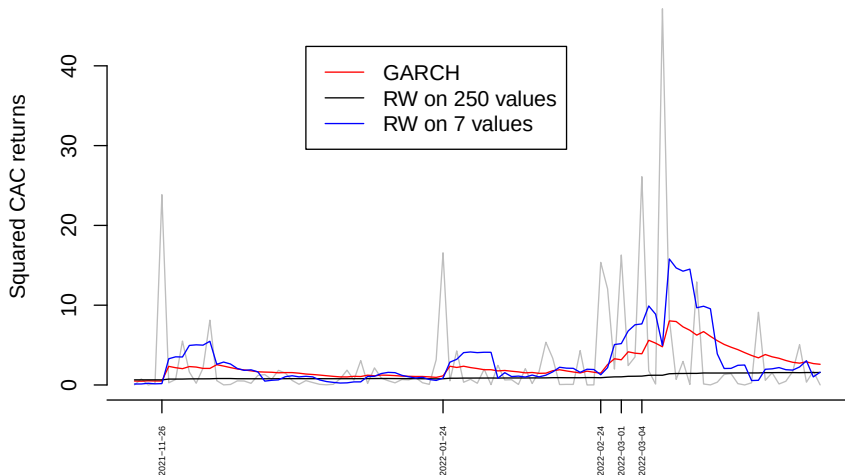
CAC returns

Volatility estimates of the CAC returns

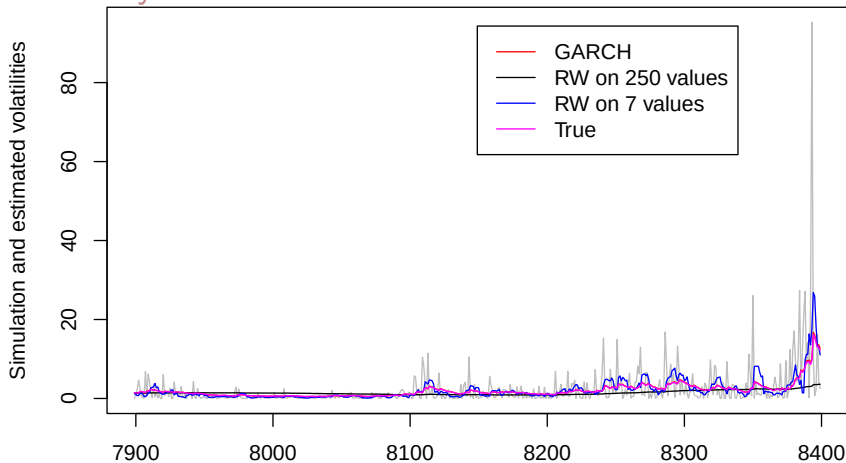
From 2021-04-26 to 2023-03-30 (GARCH estimated from 1990-03-01 to 2006-10-26)



Zoom (from 2021-11-22 to 2022-04-11)



The volatility is known on simulations



Diebold-Mariano test to compare two prediction methods

Let two methods of predictions with forecast errors e_t^A and e_t^B for $t = 1, \dots, n$. Consider a Loss $L(\cdot)$, for example the quadratic loss $L(x) = x^2$ or the L^1 loss $L(x) = |x|$. Under stationarity, let $\mu = E d_t$ with $d_t = L(e_t^A) - L(e_t^B)$. Under general assumptions, the CLT holds:

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n (d_t - \mu) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \lim_{n \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n d_t \right)$$

From a consistent estimator of the long-run variance

$$\sigma^2 = \sum_{h=-\infty}^{\infty} \text{Cov}(d_t, d_{t-h}),$$

it is obvious to test $H_0: \mu = 0$ against $H_1: \mu \neq 0$ (or $\mu > 0$ or $\mu < 0$).[§]

[§]For a race of more than two horses, see the MCS method of Hansen et al. (2003)

Diebold-Mariano test H_0 : the two methods are $\simeq$

H_1 : the alternative method is less accurate than the GARCH

```
> res1<-(rend[(nest+1):n]^2-res$sig2[(nest+1):n])
> res2<-(rend[(nest+1):n]^2-rolling.vol.long[(nest+1):n])
> res3<-(rend[(nest+1):n]^2-rolling.vol.short[(nest+1):n])
>
> sqrt(mean(res1^2)); sqrt(mean(res2^2)); sqrt(mean(res3^2))
[1] 5.798827
[1] 6.230885
[1] 6.140517
>
> dm.test(res1,res2,h=1,alternative ="less")
```

Diebold-Mariano Test

```
data:  res1res2
DM = -3.9248, Forecast horizon = 1, Loss function power = 2,
p-value= 4.41e-05
alternative hypothesis: less
```


Diebold-Mariano test H_0 : the two methods are $\simeq$

H_1 : the alternative method is less accurate than the GARCH

```
> dm.test(res1, res3, h=1, alternative = "less")
```

Diebold–Mariano Test

```
data: res1res3
```

```
DM = -4.0797, Forecast horizon = 1, Loss function power = 2,
```

```
p-value = 2.296e-05
```

```
alternative hypothesis: less
```

1 Financial series and GARCH models

2 Stationarity conditions

- Weak ARMA representation
- Stationarity of the GARCH(1,1)
- Stationarity of the GARCH(p, q)

3 Other properties

The aim is to find which parameter constraints are required for the existence of a strictly stationary (non explosive) solution to a GARCH equation.

We are looking for **non anticipative** (or causal) solution, i.e. ϵ_t is a function of $\eta_t, \eta_{t-1}, \dots$

ARMA representation of the squared returns

- If it exists, the linear innovation of ϵ_t^2 is $v_t = \epsilon_t^2 - \sigma_t^2$.
 $\implies$ semi-strong ARMA($\max(p, q), p$) form:

$$\epsilon_t^2 = \omega + \sum_{i=1}^r (\alpha_i + \beta_i) \epsilon_{t-i}^2 + v_t - \sum_{j=1}^p \beta_j v_{t-j}, \quad t \in \mathbb{Z}.$$

where $r = \max(p, q)$.

► Detailed GARCH(1,1) case

- Link with the stylized facts: uncorrelatedness of (ϵ_t) (regardless of the specification of σ_t^2), autocorrelation of (ϵ_t^2) (that of an ARMA).
- The ARMA representation is weak, and assume the existence of σ_t^2 . ⚡ One **can not** use the semi-strong ARMA representation to obtain stationarity conditions.

Strict stationarity of the GARCH(1,1)

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \omega > 0, \alpha, \beta \geq 0. \end{cases}$$

$$\begin{aligned} \sigma_t^2 &= \omega + a(\eta_{t-1})\sigma_{t-1}^2 && (a(z) = \alpha z^2 + \beta) \\ &= \omega + a(\eta_{t-1})\{\omega + a(\eta_{t-2})\sigma_{t-2}^2\} \\ &= \omega \left[1 + \sum_{n=1}^N a(\eta_{t-1}) \dots a(\eta_{t-n}) \right] + a(\eta_{t-1}) \dots a(\eta_{t-N-1})\sigma_{t-N-1}^2 \\ &:= h_t(N) + a(\eta_{t-1}) \dots a(\eta_{t-N-1})\sigma_{t-N-1}^2. \end{aligned}$$

Let $h_t = \lim_{N \rightarrow \infty} \uparrow h_t(N) \in [0, +\infty]$.

$$h_t(N) = \omega + a(\eta_{t-1})h_{t-1}(N-1) \implies h_t = \omega + a(\eta_{t-1})h_{t-1}.$$

Does the limiting process

$$h_t = \left\{ 1 + \sum_{n=1}^{\infty} a(\eta_{t-1}) \dots a(\eta_{t-n}) \right\} \omega$$

have finite values ?

► Cauchy's criterion

$$[a(\eta_{t-1}) \dots a(\eta_{t-n})]^{1/n} = \exp \left[\frac{1}{n} \sum_{i=1}^n \log\{a(\eta_{t-i})\} \right] \rightarrow e^{\gamma} \quad a.s.,$$

$$\gamma := E \log\{\alpha \eta_t^2 + \beta\} \in [-\infty, +\infty).$$

Strict stationarity

The GARCH(1,1) model has a (unique) strictly stationary solution (ergodic and non anticipative) iff

$$\gamma := E \log \{\alpha \eta_t^2 + \beta\} < 0.$$

[Nelson, 1990]

► Ergodic Theorem

ARCH(1) case ($\beta = 0$): $0 \leq \alpha < \exp\{-E(\log \eta_t^2)\}$.

If $\eta_t \sim \mathcal{N}(0, 1)$: $0 \leq \alpha < 3.56$.

Remark: By Jensen we have

$$\alpha + \beta < 1 \Rightarrow \gamma < 0, \quad \text{and} \quad \gamma < 0 \Rightarrow \beta < 1.$$

Proof of the proposition

- If $\gamma < 0$, Cauchy's criterion and the ergodic theorem entail that $\epsilon_t := \sqrt{h_t} \eta_t$ is a str. stat. (ergodic and causal) solution.
- If $\gamma > 0$, then $\sigma_t^2 = +\infty$ a.s. because $\sigma_t^2 \geq h_t(N) \forall N$ and $h_t(N) \rightarrow \infty$ when $N \rightarrow \infty$.
- If $\gamma = 0$ then $\{\sum_{i=1}^N \log a(\eta_{t-i})\}_N$ is a random walk. If it is non-degenerated then $\limsup_{n \rightarrow \infty} a(\eta_{t-1}) \cdots a(\eta_{t-n}) = +\infty$ and then $h_t(N) \rightarrow \infty$. Also true for degenerated r.w. ► Chung-Fuchs
- Let $\tilde{\epsilon}_t = \sigma_t \eta_t$ be a strictly stationary solution. We have

$$\sigma_t^2 - h_t = \{h_t(N) - h_t\} + a(\eta_{t-1}) \dots a(\eta_{t-N-1}) \sigma_{t-N-1}^2, \forall N.$$

When $\gamma < 0$, the right-hand side tends to 0 in probability when $N \rightarrow \infty$. The left-hand side (which does not depend on N) is thus equal to 0 with probability 1.

Second-order stationarity of the GARCH(1,1)

If ϵ_t is second-order stationary and causal

$$E(\epsilon_t^2) = E(\sigma_t^2)E(\eta_t^2) = E(\sigma_t^2) = \omega + (\alpha + \beta)E(\epsilon_{t-1}^2)$$

and thus

$$(1 - \alpha - \beta)E(\epsilon_t^2) = \omega.$$

It is thus necessary to have $\alpha + \beta < 1$. Moreover $E(\epsilon_t^2) > 0$.

Conversely: if $\alpha + \beta < 1$ we have $\gamma < 0$. The strict stationary solution satisfies

$$\begin{aligned} E(\epsilon_t^2) = E(h_t) &= \left[1 + \sum_{n=1}^{+\infty} E\{a(\eta_{t-1}) \dots a(\eta_{t-n})\} \right] \omega \\ &= \left[1 + \sum_{n=1}^{+\infty} \{Ea(\eta_1)\}^n \right] \omega = \frac{\omega}{1 - (\alpha + \beta)}. \end{aligned}$$

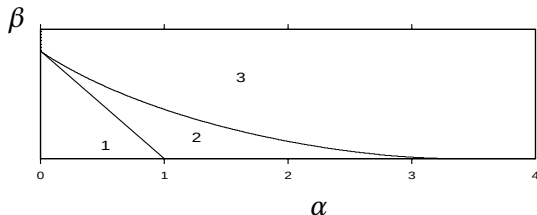
Second-order stationarity condition

The GARCH(1,1) model has a second-order and causal solution iff

$$\alpha + \beta < 1.$$

Stationarity regions of the GARCH(1,1) when $\eta_t \sim \mathcal{N}(0,1)$.

1: Second-order stationarity; 1 et 2: Strict stationarity; 3: Non stationarity.



Integrated GARCH (IGARCH)

If $\alpha + \beta = 1$ (and $\alpha\eta_t^2$ is non-degenerate) there exists a strictly stationary solution, but no second-order stationary solution. By analogy to the unit-root integrated ARMA, it is said that we have an IGARCH(1,1) model

$$\sigma_t^2 = \omega + \sigma_{t-1}^2 + u_t$$

where

$$u_t = (\alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2 - \sigma_{t-1}^2) = \sigma_{t-1}^2(\alpha\eta_{t-1}^2 + \beta - 1)$$

is centred. Since (u_t) is not iid, the analogy is misleading ...

💣 One **can not** use the semi-strong ARMA representation to obtain stationarity conditions.

Strict stationarity of the GARCH(p, q) model

Vectorial representation of $\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$:

$$\underline{z}_t = \underline{b}_t + A_t \underline{z}_{t-1}$$

where

$$\underline{z}_t = (\epsilon_t^2, \dots, \epsilon_{t-q+1}^2, \sigma_t^2, \dots, \sigma_{t-p+1}^2)' \in \mathbb{R}^{p+q},$$

$$\underline{b}_t = (\omega \eta_t^2, 0, \dots, \omega, 0, \dots, 0)' \in \mathbb{R}^{p+q},$$

$$A_t = \begin{pmatrix} \alpha_1 \eta_t^2 & \cdots & \alpha_q \eta_t^2 & \beta_1 \eta_t^2 & \cdots & \beta_p \eta_t^2 \\ & I_{q-1} & 0 & 0 & & \\ \alpha_1 & \cdots & \alpha_q & \beta_1 & \cdots & \beta_p \\ & 0 & & I_{p-1} & 0 & \end{pmatrix}.$$

► The GARCH(1,2) case

Equivalence between the univariate GARCH and the multivariate AR(1)

- $\underline{z}_t$ AR(1) solution $\Rightarrow \epsilon_t = \sqrt{\underline{z}_{q+1,t}} \eta_t$ GARCH solution.
- $(\epsilon_t) \sim \text{GARCH}(p, q) \Rightarrow \underline{z}_t = (\epsilon_t^2, \dots, \sigma_{t-p+1}^2)' \sim \text{AR}(1)$.

thus

- $(\underline{z}_t)$ strictly stationary $\Leftrightarrow (\epsilon_t)$ strictly stationary.
- $(\underline{z}_t) \in L^1 \Leftrightarrow (\epsilon_t) \in L^2$.

GARCH(p, q): strict stationarity

Iterating

$$\underline{z}_t = \underline{b}_t + A_t \underline{z}_{t-1},$$

do we obtain

$$\begin{aligned} \underline{z}_t &= \underline{b}_t + A_t \{ \underline{b}_{t-1} + A_{t-1} \underline{z}_{t-2} \} \\ &= \underline{b}_t + \sum_{k=1}^{\infty} A_t A_{t-1} \dots A_{t-k+1} \underline{b}_{t-k} \quad ? \end{aligned}$$

Pb: validity of the infinite sum.

Top Lyapounov exponent (general case)

For any strictly stationary and ergodic sequence $\mathbf{A} = (A_t)$ of random matrices such that $E \log^+ \|A_t\| < \infty$ let

$$\begin{aligned}\gamma = \gamma(\mathbf{A}) &:= \inf_{t \in \mathbb{N}^*} \frac{1}{t} E(\log \|A_t A_{t-1} \dots A_1\|) \\ &= \lim_{t \rightarrow \infty} a.s. \frac{1}{t} \log \|A_t A_{t-1} \dots A_1\|.\end{aligned}$$

- The equivalence between the two definitions of γ is shown by the subadditive ergodic theorem of Kingman (1973).
- $\gamma \leq E(\log \|A_1\|)$, with equality in dimension 1.
- If $A_t = A$ for all $t \in \mathbb{Z}$, we have $\gamma = \log \rho(A)$.[¶]
- γ is independent of the choice of the norm.
- $\lim_{t \rightarrow \infty} a.s. \|A_0 \dots A_{-t}\| = 0 \Rightarrow \gamma < 0$ (Bougerol and Picard, 1992)

[¶]Gelfand's formula: $\rho(A) = \lim_n \|A^n\|^{1/n}$

Top Lyapounov exponent associated with a GARCH(p, q)

With the norm $\|A\| = \sum |a_{ij}|$, $E \log^+ \|A_t\| \leq E \|A_t\| < \infty$.

Let

$$\underline{z}_t = \underline{b}_t + \sum_{k=1}^{\infty} A_t A_{t-1} \dots A_{t-k+1} \underline{b}_{t-k} \in [0, \infty]^{p+q}.$$

We have

$$\|\underline{z}_t\| \leq \|\underline{b}_t\| + \sum_{n=0}^{\infty} \|A_t A_{t-1} \dots A_{t-n}\| \|\underline{b}_{t-n-1}\|$$

where

$$\begin{aligned} & \|A_t \dots A_{t-n}\|^{1/n} \|\underline{b}_{t-n-1}\|^{1/n} \\ &= \exp \left[\frac{1}{n} \log \|A_t \dots A_{t-n}\| + \frac{1}{n} \log \|\underline{b}_{t-n-1}\| \right] \\ &\xrightarrow{a.s.} e^\gamma \text{ (since } E |\log \|\underline{b}_{t-n-1}\|| < \infty \text{)}. \end{aligned}$$

By Cauchy's rule, $\gamma < 0$ implies that $\underline{z}_t$ is well-defined.

Strict stationarity of a GARCH(p, q)

Strict stationarity of a GARCH(p, q)

The GARCH (p, q) model has a (unique) non anticipative strictly stationary solution iff

$$\gamma(\mathbf{A}) < 0.$$

► Proof

[Bougerol & Picard, 1992]

- In general $\gamma(\mathbf{A})$ can not be computed explicitly, except in the **GARCH(1,1)** case:

$$\gamma(\mathbf{A}) = E \log(\alpha_1 \eta_t^2 + \beta_1)$$

► Proof

- $\gamma(\mathbf{A}) < 0 \Rightarrow \sum_{j=1}^p \beta_j < 1.$
- The coefficient can be evaluated by Monte Carlo simulations.

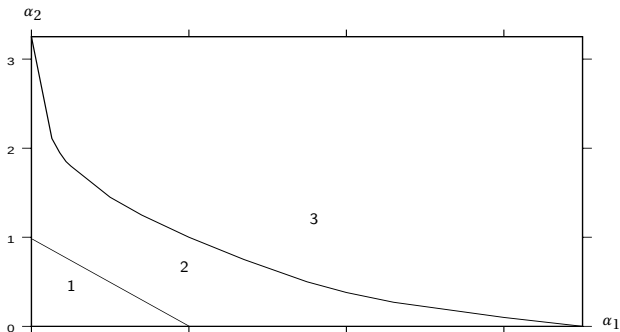
► Proof

Stationarity regions for the ARCH(2) model: $\epsilon_t = \sqrt{1 + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2} \eta_t$, $\eta_t \sim \mathcal{N}(0, 1)$.

1: Second-order stationarity;

1 et 2: Strict stationarity;

3: Non stationarity



Second-order stationarity

Second-order stationarity of a GARCH(p, q)

The GARCH (p, q) model admits a non anticipative second-order stationarity solution iff $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1$.

[► Proof](#)

The condition is equivalent to $\rho(EA_t) < 1$. It implies the strict stationarity (using Kesten and Spitzer (1984)):

$$\gamma(\mathbf{A}) \leq \gamma(EA_t) = \log \rho(EA_t) < 0.$$

Under this condition

$$\text{Var}(\epsilon_t) = \frac{\omega}{1 - \sum_{i=1}^q \alpha_i - \sum_{j=1}^p \beta_j}.$$

IGARCH model: $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j = 1$.

Properties of the marginal distribution:

Let (ϵ_t) be a strictly stationary GARCH(p, q).

Existence of $2m$ -order moments

If $E(\eta_t^{2m}) < \infty$ with $m \geq 1$, we have

$$E(\epsilon_t^{2m}) < \infty \iff \rho\{E(A_t^{\otimes m})\} = \rho\{E(A_t \otimes \cdots \otimes A_t)\} < 1.$$

► Vec and Kronecker product operators

Proof that $\rho := \rho\{E(A_t^{\otimes m})\} < 1$ implies $E(\epsilon_t^{2m}) < \infty$

We have

$$E(\epsilon_t^{2m}) < \infty \text{ iff } E\|\underline{z}_t\|^m < \infty,$$

which is true if

$$\sum_{k=1}^{\infty} \|A_t \cdots A_{t-k+1} \underline{b}_{t-k}\|_m < \infty,$$

where $\|X\|_m = (E\|X\|^m)^{1/m}$ with $\|X\| = \sum_i |x_i|$ and $\|A\| = \sum |a_{ij}|$.

Using $\|A\|\|B\| = \|A \otimes B\|$ when $a_{ij} \geq 0$ and $b_{ij} \geq 0$, and also $(Ab)^{\otimes m} = A^{\otimes m} b^{\otimes m}$ ($AB \otimes CD = (A \otimes C)(B \otimes D)$),

$$\begin{aligned} E\|A_t \cdots A_{t-k+1} \underline{b}_{t-k}\|^m &= E\|(A_t \cdots A_{t-k+1} \underline{b}_{t-k})^{\otimes m}\| \\ &= E\|A_t^{\otimes m} \cdots A_{t-k+1}^{\otimes m} \underline{b}_{t-k}^{\otimes m}\| \\ &= \|(EA_1^{\otimes m})^k E\underline{b}_{t-k}^{\otimes m}\| \leq K\rho^{*k} \end{aligned}$$

for some $\rho^* < 1$.

GARCH(1,1) example

$$A_t = (\eta_t^2, 1)'(\alpha, \beta) \Rightarrow E(A_t^{\otimes m}) = E\{(\eta_t^2, 1)^{\otimes m}\}(\alpha, \beta)^{\otimes m}.$$

Moments of order 4: $A_t^{\otimes 2} = (\eta_t^4, \eta_t^2, \eta_t^2, 1)'(\alpha^2, \alpha\beta, \alpha\beta, \beta^2)$ has 3 eigenvalues equal to 0, thus $\rho(EA_t^{\otimes 2}) = \text{Trace} = \mu_4\alpha^2 + 2\alpha\beta + \beta^2$ where $\mu_{2i} = E(\eta_t^{2i})$.

Moments of order $2m$:

$$E(\epsilon_t^{2m}) < \infty \Leftrightarrow \sum_{i=0}^m \binom{m}{i} \alpha^i \beta^{m-i} \mu_{2i} < 1.$$

Computation of the fourth-order moment

$$E(\epsilon_t^4) = \frac{\omega^2(1 + \alpha + \beta)}{(1 - \mu_4\alpha^2 - \beta^2 - 2\alpha\beta)(1 - \alpha - \beta)} \mu_4.$$

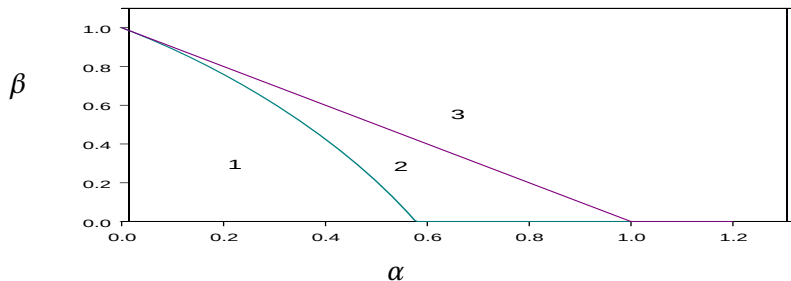
► Details

Regions for the existence of GARCH(1,1) moments.

1: Moment of order 4

1 et 2: Moment of order 2

3: Infinite variance



Kurtosis:

Kurtosis of the conditional distribution:

$$\begin{aligned} E(\epsilon_t^{2k} | \epsilon_u, u < t) &= \sigma_t^{2k} E(\eta_t^{2k}) \\ \Rightarrow \frac{E(\epsilon_t^4 | \epsilon_u, u < t)}{\{E(\epsilon_t^2 | \epsilon_u, u < t)\}^2} &= \frac{E(\eta_t^4)}{\{E(\eta_t^2)\}^2} := \kappa_\eta. \end{aligned}$$

Kurtosis of the marginal distribution:

$$\kappa_\epsilon := \frac{E(\epsilon_t^4)}{\{E(\epsilon_t^2)\}^2} = \frac{E[E(\epsilon_t^4 | \epsilon_u, u < t)]}{\{E[E(\epsilon_t^2 | \epsilon_u, u < t)]\}^2} = \frac{E(\sigma_t^4)}{\{E(\sigma_t^2)\}^2} \kappa_\eta > \kappa_\eta.$$

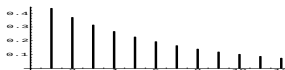
In the GARCH(1,1) case, $\kappa_\epsilon = \frac{1-(\alpha+\beta)^2}{1-(\alpha+\beta)^2-\alpha^2(\mu_4-1)} \kappa_\eta$ is an increasing function of $\frac{\alpha^2(\mu_4-1)}{1-(\alpha+\beta)^2}$.

Computing the autocovariances of ϵ_t^2

The autocovariance function can be recursively obtained from the vectorial representation.

► GARCH(1,1) case

Autocorrelations of the squares of a GARCH(1,1) model: $\epsilon_t = \sigma_t \eta_t$, $\sigma_t^2 = 1 + 0.3\epsilon_{t-1}^2 + 0.55\sigma_{t-1}^2$, $(\eta_t) \mathcal{N}(0, 1)$.



Partial autocorrelations of the same model.



Forecasting a second-order stationary GARCH(p, q)

The optimal (in L^2 sense) forecast of ϵ_t is 0. More generally, for $h \geq 0$

$$E(\epsilon_{t+h}|\epsilon_u, u < t) = E\{E(\epsilon_{t+h}|\epsilon_u, u < t+h)|\epsilon_u, u < t\} = 0, \quad t \in \mathbb{Z}.$$

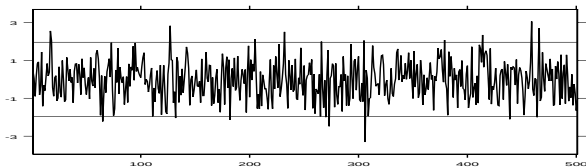
At the horizon $h+1 > 0$, the prediction of the square is obtained recursively by

$$\begin{aligned} E(\epsilon_{t+h}^2|\epsilon_u, u < t) &= E(\sigma_{t+h}^2|\epsilon_u, u < t) \\ &= \omega + \sum_{i=1}^q \alpha_i E(\epsilon_{t+h-i}^2|\epsilon_u, u < t) + \sum_{j=1}^p \beta_j E(\sigma_{t+h-j}^2|\epsilon_u, u < t), \end{aligned}$$

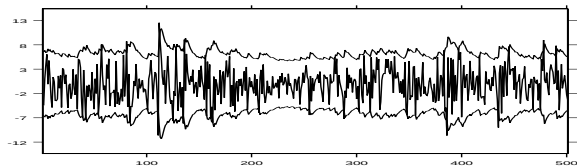
with

$$\begin{aligned} E(\epsilon_{t+h-i}^2|\epsilon_u, u < t) &= E(\sigma_{t+h-i}^2|\epsilon_u, u < t), \quad i \leq h \\ E(\epsilon_{t+h-i}^2|\epsilon_u, u < t) &= \epsilon_{t+h-i}^2, \quad i > h \\ E(\sigma_{t+h-i}^2|\epsilon_u, u < t) &= \sigma_{t+h-i}^2, \quad i \geq h. \end{aligned}$$

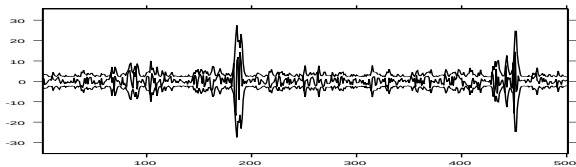
Prediction 95%-intervals at horizon 1, for a strong white noise $\mathcal{N}(0,1)$.



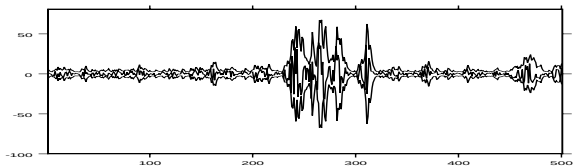
Prediction 95%-intervals at horizon 1, for the GARCH(1,1) with $\omega=1, \alpha=0.1, \beta=0.8$ and (η_t) iid $\mathcal{N}(0,1)$.



Prediction 95%-intervals at horizon 1, for the GARCH(1,1) with $\omega = 1, \alpha = 0.6, \beta = 0.2$ and (η_t) iid $\mathcal{N}(0,1)$.



Prediction 95%-intervals at horizon 1, for the IGARCH(1,1) with $\omega = 1, \alpha = 0.7, \beta = 0.3$ and (η_t) iid $\mathcal{N}(0,1)$.



Forecasting a stationary AR(1)-GARCH(1,1)

$$\begin{cases} X_t &= \phi X_{t-1} + \epsilon_t \\ \epsilon_t &= \sigma_t \eta_t \\ \sigma_t^2 &= \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases} \quad \omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1, |\phi| < 1.$$

Since $X_{t+h} = \phi^{h+1} X_{t-1} + \epsilon_{t+h} + \dots + \phi^h \epsilon_t$, we have

$E(X_{t+h} | X_u, u < t) = \phi^{h+1} X_{t-1}$, and it can be shown that

$V(X_{t+h} | X_u, u < t) = \sum_{i=0}^h \phi^{2(h-i)} \text{Var}(\epsilon_{t+i} | \epsilon_u, u < t)$

$$= \frac{\omega(1 - \phi^{2(h+1)})}{\{1 - (\alpha + \beta)\}(1 - \phi^2)} + \left\{ \sigma_t^2 - \frac{\omega}{1 - (\alpha + \beta)} \right\} \frac{\phi^{2(h+1)} - (\alpha + \beta)^{(h+1)}}{\phi^2 - (\alpha + \beta)}$$

if $\phi^2 \neq \alpha + \beta$

$$V(X_{t+h} | X_u, u < t) = \frac{\omega(1 - \phi^{2(h+1)})}{(1 - \phi^2)^2} + \left\{ \sigma_t^2 - \frac{\omega}{1 - (\alpha + \beta)} \right\} (h+1) \phi^{2h}$$

if $\phi^2 = \alpha + \beta$. (low accuracy when σ_t^2 large)

Nonstationary AR(1)-GARCH(1,1)

If $|\phi| = 1$, taking 0 as initial values for the negative dates

$$V(X_{t+h}|X_u, u < t)$$

$$= \frac{\omega h}{\{1 - (\alpha + \beta)\}} + \left\{ \sigma_t^2 - \frac{\omega}{1 - (\alpha + \beta)} \right\} \frac{1 - (\alpha + \beta)^{(h+1)}}{1 - (\alpha + \beta)}.$$

(the impact of the difference between σ_t^2 and $E\epsilon_t^2$ does not vanish as $h \rightarrow \infty$)

If $\alpha + \beta = 1$,

$$V(\epsilon_{t+h}|\epsilon_u, u < t) = \omega h + \sigma_t^2, \quad \text{for all } h \geq 0.$$

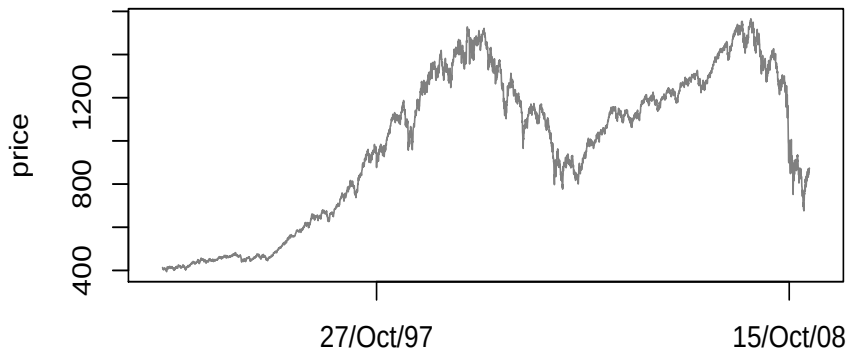
(the impact of σ_t^2 on the prediction accuracy remains constant as $h \rightarrow \infty$)

Fin du chapitre 1



Stylized Facts (Mandelbrot (1963))

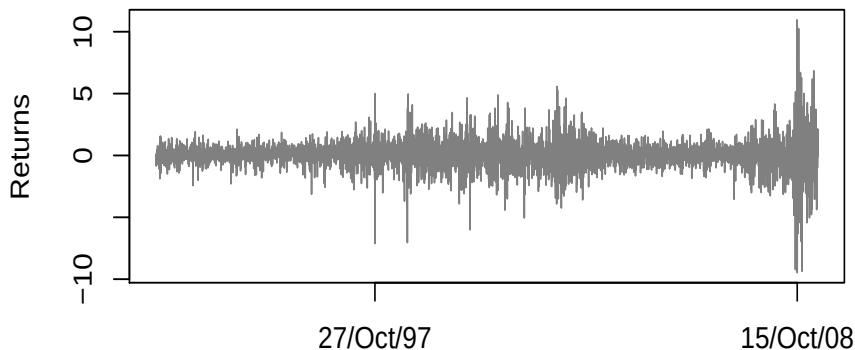
Non stationarity of the prices



S&P 500, from March 2, 1992 to April 30, 2009

Stylized Facts

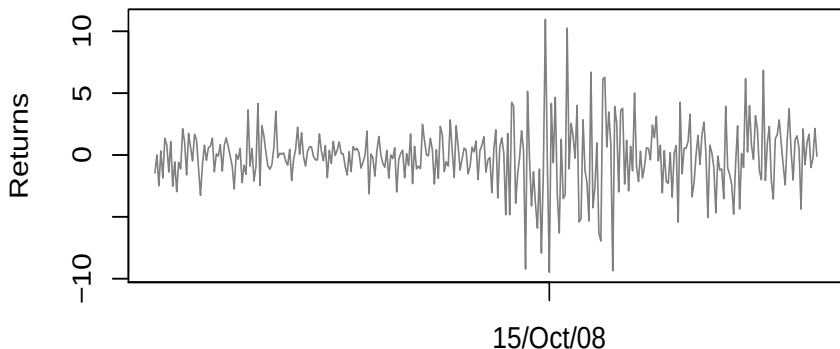
Possible stationarity of the returns



S&P 500 returns, from March 2, 1992 to April 30, 2009

Stylized Facts

Conditional heteroskedasticity (compatible with marginal homoscedasticity and even stationarity)

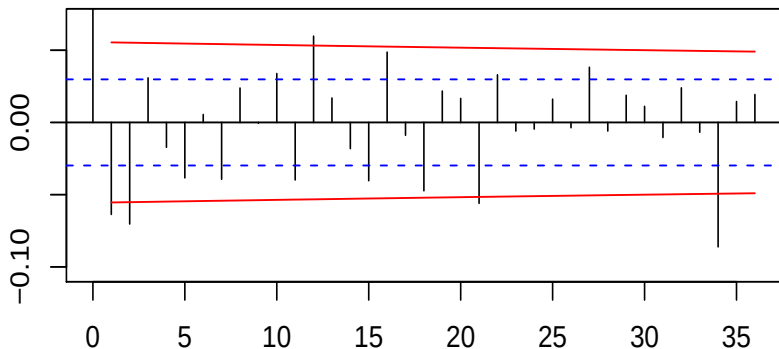


Stylized Facts

Dependence without correlation (see FZ 2009)

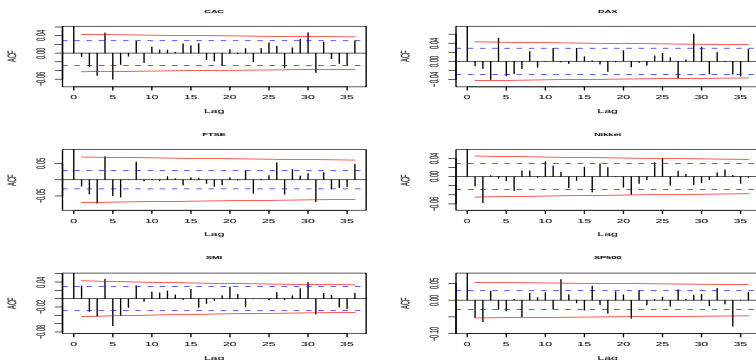
<http://perso.univ-lille3.fr/~cfrancq/Christian-Francq/Generalized-Bartlett-Formula.html>

for the interpretation of the red lines)



Stylized Facts

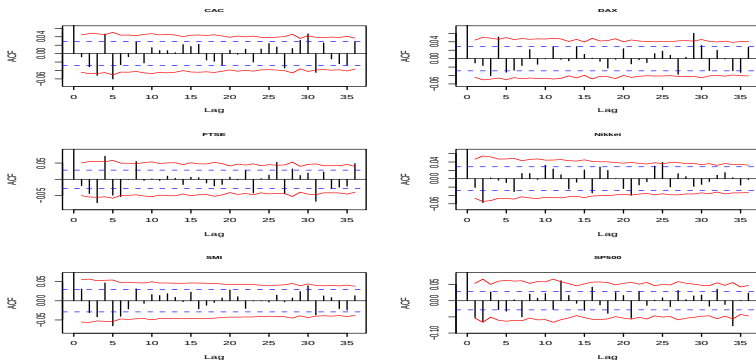
Dependence without correlation (significance bands under the GARCH(1,1) assumption)



Empirical autocorrelations of daily stock returns

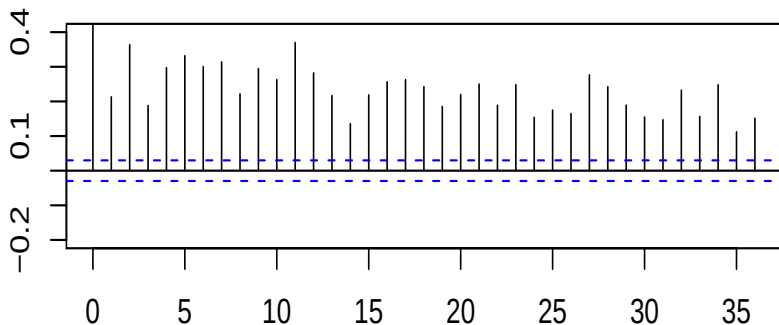
Stylized Facts

Dependence without correlation (the significance bands in red are estimated nonparametrically)



Stylized Facts

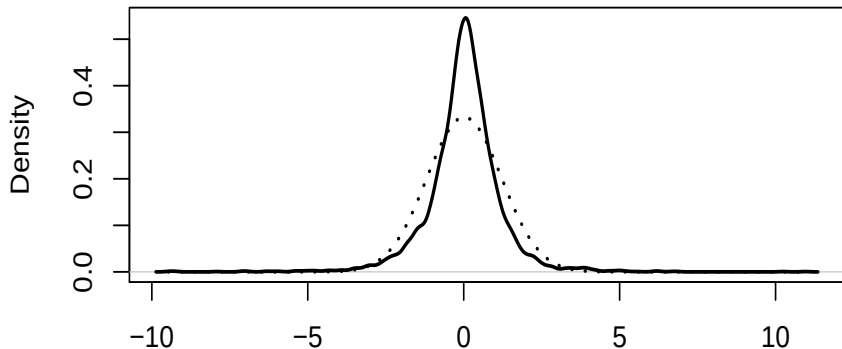
Correlation of the squares



Autocorrelations of the squares of the S&P 500 returns

Stylized Facts

Tail heaviness of the distributions



◀ Return

Density estimator for the S&P 500 returns (normal in dotted line)

Stylized Facts

Decreases of prices have a higher impact on the future volatility than increases of the same magnitude

Table: Autocorrelations of transformations of the CAC returns ϵ

h	1	2	3	4	5	6
$\hat{\rho}(\epsilon_{t-h}^+, \epsilon_t)$	0.03	0.07	0.07	0.08	0.08	0.12
$\hat{\rho}(-\epsilon_{t-h}^-, \epsilon_t)$	0.18	0.20	0.22	0.18	0.21	0.15

► SP 500

◀ Return

Stylized Facts

Decreases of prices have a higher impact on the future volatility than increases of the same magnitude

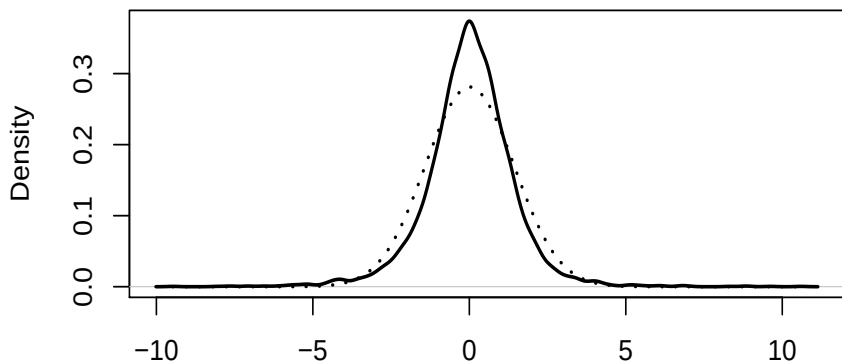
Table: Autocorrelations of transformations of the S&P 500 returns ϵ

h	1	2	3	4	5	6
$\hat{\rho}_{\epsilon}(h)$	-0.06	-0.07	0.03	-0.02	-0.04	0.01
$\hat{\rho}_{ \epsilon }(h)$	0.26	0.34	0.29	0.32	0.36	0.32
$\hat{\rho}(\epsilon_{t-h}^+, \epsilon_t)$	0.06	0.12	0.11	0.14	0.15	0.16
$\hat{\rho}(-\epsilon_{t-h}^-, \epsilon_t)$	0.25	0.28	0.23	0.24	0.28	0.23

[◀ Return](#)

Stylized Facts

Tail heaviness of the distributions



[Return](#)

Density estimator for the CAC returns (normal in dotted line)

Strict stationarity

A series (X_t) is said to be **stationary** if its probabilistic properties are the same as those of (X_{t+h}) , for any integer h .

Definition of the strict stationarity

(X_t) is **strictly stationary** if the distributions of

$$(X_1, X_2, \dots, X_k) \text{ and } (X_{1+h}, X_{2+h}, \dots, X_{k+h})$$

are the same, for all h and all $k \geq 1$.

Second-order stationarity

Mean and covariance

Let (X_t) such that $EX_t^2 < \infty$. The mean function of (X_t) is

$$\mu_X(t) = E(X_t)$$

The covariance function of (X_t) is

$$\gamma_X(r, s) = \text{Cov}(X_r, X_s)$$

Definition of the second-order stationarity

(X_t) is second-order (or weakly, or of order 2) stationary if

- (i) $\mu_X(t)$ is independent of t , and
- (ii) $\gamma_X(t, t+h)$ is independent of t , for all h .

Autocovariance and autocorrelation functions

Definition of the autocorrelations

Let (X_t) be a stationary series (of order 2). The **autocovariance function** of (X_t) is defined by

$$\gamma_X(h) = \text{Cov}(X_t, X_{t-h}), \quad h = 0, \pm 1, \pm 2, \dots$$

The **autocorrelation function** of (X_t) is defined by

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)} = \text{Cor}(X_t, X_{t+h}), \quad h = 0, \pm 1, \pm 2, \dots$$

The parameter h is called the lag or delay.

Remark: The autocovariance is an even function

$$\gamma_X(h) = \gamma_X(-h), \quad \rho_X(h) = \rho_X(-h).$$

Link between strict stationarity and stationarity of order 2

- (X_t) strictly stationary and $EX_1^2 < \infty \Rightarrow (X_t)$ second-order stationary.
- If (X_t) is Gaussian^{||}, then second-order stationary $\Leftrightarrow (X_t)$ strictly stationary.

Remark

In Finance, stationarity of order 2 is often considered as **more restrictive** than the strict stationarity, because the financial returns often display distributions with fat tails (and thus only low-order moments are likely to exist).

◀ Return

^{||}all linear combination of the X_t 's is normally distributed

White noise

Definition of a weak white noise

A (weak) white noise is a second-order stationary sequence (ϵ_t) of centred and uncorrelated random variable:

$$E(\epsilon_t) = 0, \quad \text{Var}(\epsilon_t) = \sigma^2, \quad \text{Cov}(\epsilon_t, \epsilon_{t'}) = 0, \quad t \neq t'$$

Notation: $(\epsilon_t) \sim WN(0, \sigma^2)$.

Autocorrelation function

$$\gamma_\epsilon(h) = \begin{cases} \sigma^2, & h = 0 \\ 0, & h \neq 0. \end{cases}$$

Strong and semi-strong white noise

Definition of a strong white noise

A **strong white noise** is a sequence (ϵ_t) of independent and identically distributed (iid) variables, with zero mean finite variance σ^2 .

Notation: $(\epsilon_t) \text{ iid } (0, \sigma^2)$.

Definition of a semi-strong white noise

A **semi-strong white noise** is a strictly and second-order stationary sequence (ϵ_t) such that 0 is the optimal prediction of ϵ_t given its past.

Notation: $(\epsilon_t) \text{ MD } (0, \sigma^2)$ (MD pour martingale difference).

Link between the types of white noises

weak WN $\supset$ semi-strong $\supset$ strong $\supset$ Gaussian WN

Remarks:

- In the sequel we denote by (ϵ_t) a weak or semi-strong white noise, and by (η_t) a strong white noise.
- The white noise plays an important role in time series, as error terms for more sophisticated models.

◀ Return

ARMA model

The ARMA models take into account the **second-order** structure of a stationary series (X_t) : expectation, variance, autocorrelations.

ARMA(p, q):

$$\left\{ \begin{array}{l} X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t - \psi_1 \epsilon_{t-1} - \cdots - \psi_q \epsilon_{t-q} \\ (\epsilon_t) \text{ weak white noise} \end{array} \right.$$

Characterization: Under the stationarity assumption, it can be shown (see Brockwell and Davis, 1991) that

$$\text{ARMA}(p, q) \iff \gamma_X(h) - \sum_{i=1}^p \phi_i \gamma_X(h-i) = 0, \quad \forall h > q.$$

Example of application for the characterization

Let the MA(2)

$$X_t = \epsilon_t - \alpha\epsilon_{t-1} - \beta\epsilon_{t-2}.$$

Which model is satisfied by the aggregated process $Y_t = X_{2t}$?

Example of application for the characterization

Let the MA(2)

$$X_t = \epsilon_t - \alpha\epsilon_{t-1} - \beta\epsilon_{t-2}.$$

Which model is satisfied by $Y_t = X_{2t}$?

Solution: We have

$$\begin{aligned}\gamma_Y(1) &= \text{cov}(\epsilon_{2t} - \alpha\epsilon_{2t-1} - \beta\epsilon_{2t-2}, \epsilon_{2t-2} - \alpha\epsilon_{2t-3} - \beta\epsilon_{2t-4}) \\ &= -\beta\sigma^2\end{aligned}$$

and

$$\gamma_Y(h) = 0 \quad \forall h > 1,$$

therefore (Y_t) follows a MA(1) of the form $Y_t = u_t - \theta u_{t-1}$, with θ such that

$$\rho_Y(1) = \frac{-\theta}{1+\theta^2} = \frac{-\beta}{1+\alpha^2+\beta^2}.$$

Theoretical forecast of X_t such that $EX_t^2 < \infty$

Denote by $\sigma(X_u, u < t)$ the past of X_t and by $\mathcal{H}_X(t-1)$ the linear past X_t (Hilbert subspaces of L^2). **2 concepts:**

- **Conditional expectation:**

$$E(X_t | X_u, u < t) = \arg \min_{Z \in \sigma(X_u, u < t)} E(X_t - Z)^2$$

"Best" approximation of X_t by a function of its past.

- **Linear conditional expectation:**

$$EL(X_t | X_u, u < t) = \arg \min_{Z \in \mathcal{H}_X(t-1)} E(X_t - Z)^2$$

"Best" approximation of X_t by a **linear** function of its past.

Innovations and white noises

2 concepts:

- Strong innovation:

$$\epsilon_t = X_t - E(X_t | X_u, u < t)$$

is a **semi-strong white noise** (if (X_t) strictly and second-order stationary), "orthogonal" to any function of the past of X_t

$$E(\epsilon_t Z_{t-1}) = 0, \quad \forall Z_{t-1} \in \sigma(X_u, u < t).$$

- Linear innovation:

$$\epsilon_t = X_t - EL(X_t | X_{t-1}, X_{t-2}, \dots)$$

is a **weak white noise**, orthogonal to any linear function of the past

$$E(\epsilon_t Z_{t-1}) = 0, \quad \forall Z_{t-1} \in \mathcal{H}_X(t-1).$$

ARMA innovations

The $\text{ARMA}(p, q)$ satisfies

$$\phi(L)X_t = \psi(L)\epsilon_t$$

with the **lag (or backward) operator** $L^k X_t = X_{t-k}$ and

$$\phi(x) = 1 - \sum_{i=1}^p \phi_i x^i, \quad \psi(x) = 1 - \sum_{i=1}^q \psi_i x^i.$$

If (see Brockwell and Davis, 1991)

$$\phi(x)\psi(x) = 0 \Rightarrow |x| > 1$$

then $X_t = \phi^{-1}(L)\psi(L)\epsilon_t \in \mathcal{H}_\epsilon(t)$, $\epsilon_t = \psi^{-1}(L)\phi(L)X_t \in \mathcal{H}_X(t)$.

The pasts and linear **pasts of X_t and ϵ_t coincide**. The noise ϵ_t is the linear innovation (strong innovation if the noise is strong).

Stationary ergodic processes

A strictly stationary series is said to be ergodic if it satisfies the strong law of large numbers (even if the usual iid assumption is not satisfied).

Definition

A **strictly stationary** process $(Z_t)_{t \in \mathbb{Z}}$ is said to be **ergodic** if and only if, for all integer k and all Borel set B (of a suitable set)

$$n^{-1} \sum_{t=1}^n I_B(Z_t, Z_{t+1}, \dots, Z_{t+k}) \rightarrow P\{(Z_1, \dots, Z_{1+k}) \in B\} \text{ a.s.}$$

For complements and extensions, see e.g. Billingsley (1995) "Probability and Measure", Wiley.

Ergodic theorem

Any fixed transformation of a (strictly) stationary and ergodic series remains stationary and ergodic.

Theorem

If $(Z_t)_{t \in \mathbb{Z}}$ is a strictly stationary and ergodic sequence and if $(Y_t)_{t \in \mathbb{Z}}$ is defined by

$$Y_t = f(Z_t, Z_{t+1}, \dots; Z_{t-1}, Z_{t-2}, \dots),$$

where f is a measurable function from $\mathbb{R}^\infty$ to $\mathbb{R}$, then $(Y_t)_{t \in \mathbb{Z}}$ is also a strictly stationary and ergodic sequence.

Examples and counterexamples of stationary ergodic processes

- A strong white noise (η_t) is stationary and ergodic.
- An MA(q) $X_t = \sum_{i=0}^q c_i \eta_{t-i}$ or the stationary solution of the AR(1) $X_t = aX_{t-1} + \eta_t$ with $|a| \neq 1$ are stationary and ergodic.
- The process defined by

$$X_t = X, \quad \forall t,$$

where X is a non-degenerated random variable, is stationary but not ergodic.

- The process defined by

$$Y_t = X_t - X_1, \quad \forall t,$$

where (X_t) is stationary and ergodic, is not stationary and ergodic.

Martingale

In a fair and purely random game between two gamblers, the wealth of one player is a martingale.

Definition

Let $(Y_t)_{t \in \mathbb{N}}$ a sequence of real random variables and $(\mathcal{F}_t)_{t \in \mathbb{N}}$ a sequence of sigma-fields. The sequence $(Y_t, \mathcal{F}_t)_{t \in \mathbb{N}}$ is a **martingale** if and only if

- ① $\mathcal{F}_t \subset \mathcal{F}_{t+1}$;
- ② Y_t is $\mathcal{F}_t$ -measurable;
- ③ $E|Y_t| < \infty$;
- ④ $E(Y_{t+1} | \mathcal{F}_t) = Y_t$.

When $(Y_t)_{t \in \mathbb{N}}$ is said to be a martingale, one implicitly assume that $\mathcal{F}_t = \sigma(Y_u, u \leq t)$.

Martingale increments

In a fair and purely random game between two gamblers, the wealth of one player is a martingale.

Definition

Let $(\epsilon_t)_{t \in \mathbb{N}}$ a sequence of real random variables and $(\mathcal{F}_t)_{t \in \mathbb{N}}$ a sequence of sigma-fields. The sequence $(\epsilon_t, \mathcal{F}_t)_{t \in \mathbb{N}}$ is a **martingale difference** (or a martingale increment sequence) if and only if

- ① $\mathcal{F}_t \subset \mathcal{F}_{t+1}$;
- ② ϵ_t is $\mathcal{F}_t$ -measurable;
- ③ $E|\epsilon_t| < \infty$;
- ④ $E(\epsilon_{t+1} | \mathcal{F}_t) = 0$.

Examples of martingale difference

- If $(Y_t)_{t \in \mathbb{N}}$ is a martingale, then $\epsilon_t = Y_t - Y_{t-1}$ is a martingale difference.
- If $(\epsilon_t)_{t \in \mathbb{N}}$ is a martingale difference then $X_t = \sum_{i=0}^t \epsilon_i$ is a martingale.
- A semi-strong white noise is a martingale difference.
- If (X_t) is a **causal** AR(1)

$$X_t = aX_{t-1} + \eta_t, \quad |a| < 1,$$

then $\{\eta_t X_{t-1}, \sigma(\eta_u, u \leq t)\}$ is a martingale difference.

CLT for triangular series

Lindeberg's CLT

Assume that, $\forall n > 0$, $(\epsilon_{nk}, \mathcal{F}_{nk})_{k \in \mathbb{N}}$ is a square-integrable martingale difference. Let $\sigma_{nk}^2 = E(\epsilon_{nk}^2 | \mathcal{F}_{n(k-1)})$. If

$$\sum_{k=1}^n \sigma_{nk}^2 \rightarrow \sigma_0^2 \text{ en probabilité when } n \rightarrow \infty,$$

where $\sigma_0 > 0$, and

$$\forall \varepsilon > 0, \quad \sum_{k=1}^n E \epsilon_{nk}^2 1_{\{|\epsilon_{nk}| \geq \varepsilon\}} \rightarrow 0 \text{ when } n \rightarrow \infty,$$

then $\sum_{k=1}^n \epsilon_{nk} \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma_0^2)$

The Lindeberg CLT contains the usual CLT

Let $Z_1, \dots, Z_n$ be an iid sequence with finite variance,

$$\epsilon_{nk} = \frac{Z_k - EZ_k}{\sqrt{n}} \text{ and } \mathcal{F}_{nk} = \sigma(Z_1, \dots, Z_k).$$

We have $\sigma_{nk}^2 = E\epsilon_{nk}^2 = n^{-1}\text{Var}(Z_0)$ and $\sigma_0^2 = \text{Var}(Z_0)$. Moreover

$$\begin{aligned} \sum_{k=1}^n E\epsilon_{nk}^2 1_{\{|\epsilon_{nk}| \geq \varepsilon\}} &= \sum_{k=1}^n n^{-1} \int_{\{|Z_k - EZ_k| \geq \sqrt{n}\varepsilon\}} |Z_k - EZ_k|^2 dP \\ &= \int_{\{|Z_1 - EZ_1| \geq \sqrt{n}\varepsilon\}} |Z_1 - EZ_1|^2 dP \rightarrow 0 \end{aligned}$$

because $\{|Z_1 - EZ_1| \geq \sqrt{n}\varepsilon\} \downarrow \emptyset$ and $\int_{\Omega} |Z_1 - EZ_1|^2 dP < \infty$ (using the dominated convergence theorem). The condition of Lindeberg is thus satisfied.

Application of the ergodic and Lindeberg theorems

If (X_t) is a strong causal AR(1)

$$X_t = aX_{t-1} + \eta_t, \quad |a| < 1,$$

then

$$\hat{a} := \frac{\sum_{t=2}^n X_t X_{t-1}}{\sum_{t=2}^n X_{t-1}^2} \rightarrow a \quad \text{a.s.}$$

by the ergodic theorem, and

$$\sqrt{n}\{\hat{a} - a\} = \frac{n^{-1/2} \sum_{t=2}^n \eta_t X_{t-1}}{n^{-1} \sum_{t=2}^n X_{t-1}^2} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1 - a^2)$$

by Lindeberg's CLT.

Cauchy's rule

Cauchy's criterion for the convergence of a series with terms $a_n \geq 0$.

Let $\lambda = \limsup a_n^{1/n}$. We have

$$\lambda < 1 \Rightarrow \sum_{n=1}^{\infty} a_n < \infty$$

and

$$\lambda > 1 \Rightarrow \sum_{n=1}^{\infty} a_n = +\infty.$$

◀ Return

Chung-Fuchs's theorem

Theorem on random walk

If $X_1, \dots, X_n$ is iid with $EX_1 = 0$ and $E|X_1| > 0$, then a.s.

$$\limsup_{n \rightarrow \infty} \sum_{i=1}^n X_i = +\infty$$

and

$$\liminf_{n \rightarrow \infty} \sum_{i=1}^n X_i = -\infty.$$

[◀ Return](#)

Proof that $\gamma < 0$ is N and S for strict stationarity

- We have seen that $\gamma < 0$ entails a strict stationary solution $\underline{z}_t$ for the VAR(1), and thus for the GARCH(p, q) model.
- Assume that there exists a strictly stationary solution $\underline{z}_t$. We have

$$\underline{z}_0 = \underline{b}_0 + \sum_{k=0}^{t-1} A_0 \dots A_{-k} \underline{b}_{-k-1} + A_0 \dots A_{-t} \underline{z}_{-t-1} \geq \sum_{k=0}^{t-1} A_0 \dots A_{-k} \underline{b}_{-k-1},$$

thus $\|A_0 \dots A_{-k} \underline{b}_{-k-1}\| \rightarrow 0$ a.s. as $k \rightarrow \infty$. It can be shown that we then have $\|A_0 \dots A_{-k}\| \rightarrow 0$ a.s. which implies $\gamma < 0$ by a result of Bougerol and Picard (1992).

Computation of γ for a GARCH(1,1)

For the GARCH(1,1) model we have $\underline{z}_t = \underline{b}_t + A_t \underline{z}_{t-1}$ with

$$\underline{z}_t = \begin{pmatrix} \epsilon_t^2 \\ \sigma_t^2 \end{pmatrix}, \quad A_t = \begin{pmatrix} \alpha \eta_t^2 & \beta \eta_t^2 \\ \alpha & \beta \end{pmatrix} = \begin{pmatrix} \eta_t^2 \\ 1 \end{pmatrix} \begin{pmatrix} \alpha & \beta \end{pmatrix}.$$

Therefore

$$A_t \cdots A_{t-n} = \begin{pmatrix} \eta_t^2 \\ 1 \end{pmatrix} \prod_{i=1}^n a(\eta_{t-i}) \begin{pmatrix} \alpha & \beta \end{pmatrix}$$

and

$$\frac{1}{n} \log \|A_t \cdots A_{t-n}\| \rightarrow E \log a(\eta_1) \quad a.s.$$

$\sum_{j=1}^p \beta_j < 1$ is necessary for strict stationarity

$$A_t = \begin{pmatrix} \alpha_1 \eta_t^2 & \cdots & \alpha_q \eta_t^2 & \beta_1 \eta_t^2 & \cdots & \beta_p \eta_t^2 \\ & I_{q-1} & 0 & 0 & & \\ \alpha_1 & \cdots & \alpha_q & \beta_1 & \cdots & \beta_p \\ & 0 & & & I_{p-1} & 0 \end{pmatrix}, \quad B = \begin{pmatrix} \beta_1 & \cdots & \beta_p \\ & I_{p-1} & 0 \end{pmatrix}.$$

We thus have $\gamma(\mathbf{A}) \geq \gamma(\mathbf{B}) = \log \rho(B)$ and

$$\det(B - \lambda I_p) = (-1)^p (\lambda^p - \sum_{j=1}^p \beta_j \lambda^{p-j}) = (-1)^p \lambda^p \mathcal{B}(1/\lambda),$$

► first equality

with $\mathcal{B}(z) = 1 - \sum_{j=1}^p \beta_j z^j$. It is thus sufficient to show that

$$\{\mathcal{B}(z) = 0 \Rightarrow |z| > 1\} \quad \Leftrightarrow \quad \sum_{j=1}^p \beta_j < 1.$$

$\sum_{j=1}^p \beta_j < 1$ is necessary for strict stationarity (continued)

It is thus sufficient to show that

$$\left\{ \mathcal{B}(z) := 1 - \sum_{j=1}^p \beta_j z^j = 0 \Rightarrow |z| > 1 \right\} \Leftrightarrow \sum_{j=1}^p \beta_j < 1.$$

$\Rightarrow$ Assuming $\sum_{j=1}^p \beta_j \geq 1$ we show that $\mathcal{B}(z_0) = 0$ for some $|z_0| \leq 1$.

We have $\mathcal{B}(0) = 1$ and $\mathcal{B}(1) = 1 - \sum_{j=1}^p \beta_j \leq 0$. By continuity, there exists a root in $(0, 1]$.

$\Leftarrow$ If $\sum_{j=1}^p \beta_j < 1$ and $\mathcal{B}(z_0) = 0$ with $|z_0| \leq 1$, then

$$1 = \left| \sum_{j=1}^p \beta_j z_0^j \right| \leq \sum_{j=1}^p \beta_j |z_0^j| \leq \sum_{j=1}^p \beta_j$$

which is impossible.

Proof that $\sum \alpha_i + \beta_i < 1$ is NS for second-order stationarity

- If $E\epsilon_t^2 < \infty$ then

$$E\epsilon_t^2 = E\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i E\epsilon_t^2 + \sum_{j=1}^p \beta_j E\epsilon_t^2$$

which implies $1 - \sum \alpha_i + \beta_i > 0$.

- Assume $\sum \alpha_i + \beta_i < 1$. It is sufficient to show that

$$\underline{z}_t = \underline{b}_t + \sum_{k=0}^{\infty} A_t \dots A_{t-k} \underline{b}_{t-k-1} \in L^1.$$

By Cauchy's rule, this is true if

$$\lim_{k \rightarrow \infty} \|EA_t \dots A_{t-k}\|^{1/k} = \lim_{k \rightarrow \infty} \|A^k\|^{1/k} = \rho(A) < 1,$$

where $A = EA_t$.

$\sum \alpha_i + \beta_i < 1$ is NS for second-order stationarity (continued)

We have

$$A = \begin{pmatrix} \alpha_1 & \cdots & \alpha_q & \beta_1 & \cdots & \beta_p \\ & I_{q-1} & 0 & & 0 & \\ \alpha_1 & \cdots & \alpha_q & \beta_1 & \cdots & \beta_p \\ & 0 & & & I_{p-1} & 0 \end{pmatrix}$$

and it can be shown recursively that

$$\rho(A) < 1 \Leftrightarrow \sum \alpha_i + \beta_i < 1.$$

◀ Return

Autocorrelations of a GARCH(1,1)

Let

$$\begin{cases} \epsilon_t &= \sigma_t \eta_t \\ \sigma_t^2 &= \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \end{cases}$$

with η_t iid $\mathcal{N}(0,1)$ and $1 - 3\alpha^2 - \beta^2 - 2\alpha\beta > 0$. From the ARMA(1,1) representation of the squared returns

$$\rho_{\epsilon^2}(h) = (\alpha + \beta) \rho_{\epsilon^2}(h-1) = (\alpha + \beta)^h \rho_{\epsilon^2}(1), \quad h > 1.$$

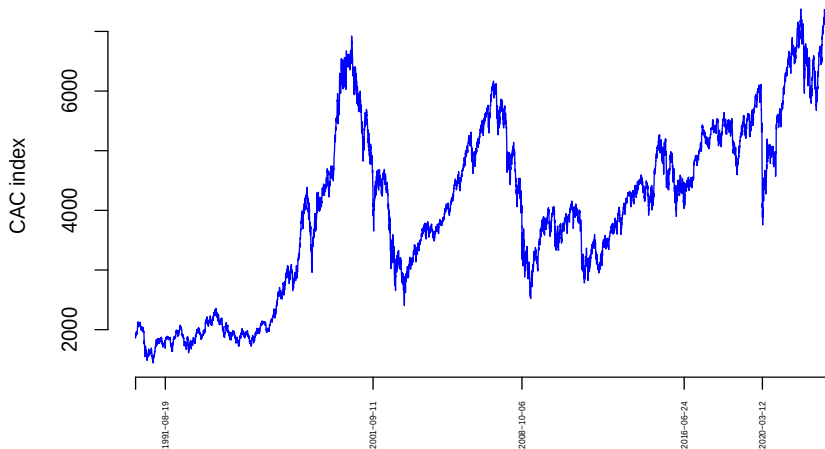
Using the MA(∞) representation

$$\epsilon_t^2 = \frac{\omega}{1 - \alpha - \beta} + v_t + \alpha \sum_{i=1}^{\infty} (\alpha + \beta)^{i-1} v_{t-i}$$

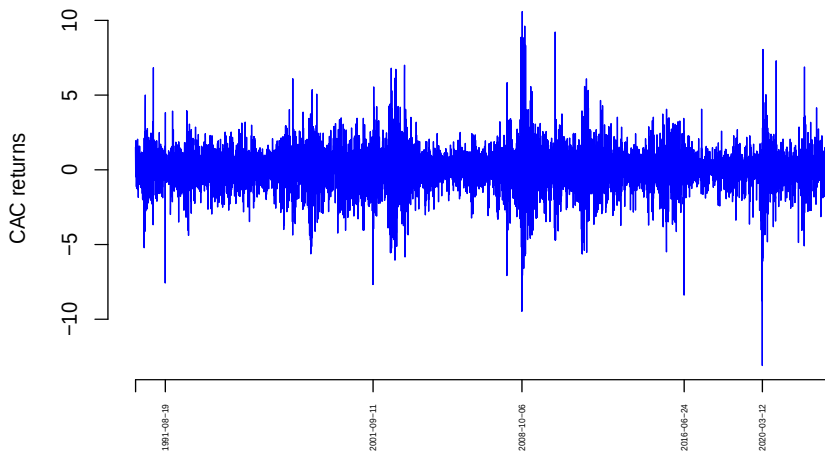
we obtain

$$\gamma_{\epsilon^2}(0) = E v_1^2 \left(1 + \frac{\alpha^2}{1 - (\alpha + \beta)^2} \right), \quad \gamma_{\epsilon^2}(1) = E v_1^2 \left(\alpha + \frac{\alpha^2(\alpha + \beta)}{1 - (\alpha + \beta)^2} \right)$$

CAC daily index from 1990-03-01 to 2023-03-30



Log-returns $\epsilon_t = \log(p_t / p_{t-1})$



◀ Return

Weak ARMA(1,1) representation of a GARCH(1,1)

Let the GARCH(1,1) model

$$\begin{cases} \epsilon_t = \sigma_t \eta_t, & \eta_t \text{ iid } (0, 1), \\ \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}$$

and **assume** that $\sigma_t^2 \in \mathcal{F}_{t-1} := \sigma(\epsilon_u, u < t)$.

Letting $v_t = \epsilon_t^2 - \sigma_t^2$, we have the **weak ARMA** representation

$$\epsilon_t^2 = v_t + \sigma_t^2 = v_t + \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 = v_t + \omega + (\alpha + \beta) \epsilon_{t-1}^2 - \beta v_{t-1}^2$$

where

$$v_t = (\eta_t^2 - 1) \sigma_t^2$$

is a weak (semi-strong) white noise.

Markovian representation of the GARCH(1,2)

For example, consider the GARCH(1,2) volatility

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2 + \beta \sigma_{t-1}^2.$$

We have the Markovian representation

$$\begin{pmatrix} \epsilon_t^2 \\ \epsilon_{t-1}^2 \\ \sigma_t^2 \end{pmatrix} = \begin{pmatrix} \omega \eta_t^2 \\ 0 \\ \omega \end{pmatrix} + \begin{pmatrix} \alpha_1 \eta_t^2 & \alpha_2 \eta_t^2 & \beta \eta_t^2 \\ 1 & 0 & 0 \\ \alpha_1 & \alpha_2 & \beta \end{pmatrix} \begin{pmatrix} \epsilon_{t-1}^2 \\ \epsilon_{t-2}^2 \\ \sigma_{t-1}^2 \end{pmatrix}.$$

[◀ Return](#)

Vec operator and Kronecker product

For a matrix A , the vector $\text{vec } A$ is obtained by stacking the columns of A .

For instance,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \text{vec}(A) = \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}$$

The Kronecker product of 2 matrices is defined by

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1d_2}B \\ \vdots & & \\ a_{d_1 1}B & \cdots & a_{d_1 d_2}B \end{pmatrix}.$$

Properties of the operators vec and $\otimes$

Assuming that the dimensions are compatible,

- $A \otimes B \neq B \otimes A$ in general,
- $(A \otimes B)' = A' \otimes B'$,
- $A \otimes (B + C) = A \otimes B + A \otimes C$,
- if A and B invertible $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$,
- $(A \otimes B)(C \otimes D) = AC \otimes BD$,
- the eigenvalues of $A \otimes B$ are the products of those of A and B ,
- $\text{vec}(A + B) = \text{vec } A + \text{vec } B$,
- $\text{vec } ABC = (C' \otimes A) \text{vec } B$,
- $\text{vec } AB = (I \otimes A) \text{vec } B = (B' \otimes I) \text{vec } A$,
- $(\text{vec } B')' \text{vec } A = \text{tr } AB = \text{tr } BA = (\text{vec } A')' \text{vec } B$,
- $\text{vec } ab' = b \otimes a$ when a and b are 2 vectors.

Computation of $\rho(B)$

$$B = \begin{pmatrix} \beta_1 & \beta_2 & \cdots & \beta_p \\ 1 & 0 & & 0 \\ 0 & \ddots & & \\ 0 & \cdots & 1 & 0 \end{pmatrix}.$$

The relation

$$\det(B - \lambda I_p) = (-1)^p (\lambda^p - \sum_{j=1}^p \beta_j \lambda^{p-j})$$

is true for $p = 2$. In this case the determinant is indeed $\lambda^2 - \lambda\beta_1 - \beta_2$. Argue by induction on p and assume that it is true up to $p - 1$.

Computation of $\rho(B)$

Computing the determinant with respect to the last column, we have

$$\begin{aligned}\det(B - \lambda I_p) &= \begin{vmatrix} \beta_1 - \lambda & \beta_2 & \cdots & \beta_p \\ 1 & -\lambda & & 0 \\ 0 & \ddots & & \\ 0 & \cdots & 1 & -\lambda \end{vmatrix} \\ &= (-1)^p \beta_p - \lambda (-1)^{p-1} \left(\lambda^{p-1} - \sum_{j=1}^{p-1} \beta_j \lambda^{p-1-j} \right) \\ &= (-1)^p \left(\lambda^p - \sum_{j=1}^p \beta_j \lambda^{p-j} \right)\end{aligned}$$

and the conclusion follows.

[◀ Return](#)

Fourth-order moment of the GARCH(1,1)

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

We have

$$\begin{aligned} E(\epsilon_t^4) &= \mu_4 E(\omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2)^2 \\ &= \mu_4 (\omega^2 + \alpha^2 E\epsilon_t^4 + \beta^2 E\sigma_t^4 + 2\omega(\alpha + \beta)E\sigma_t^2 + 2\alpha\beta E\sigma_t^4) \\ &= \frac{\mu_4 (\omega^2 + 2\omega(\alpha + \beta)E\sigma_t^2)}{1 - \mu_4 \alpha^2 - \beta^2 - 2\alpha\beta} \\ &= \frac{\mu_4 \omega^2 \left(1 + 2 \frac{\alpha + \beta}{1 - \alpha - \beta}\right)}{1 - \mu_4 \alpha^2 - \beta^2 - 2\alpha\beta} \end{aligned}$$

provided $\mu_4 \alpha^2 + \beta^2 + 2\alpha\beta < 1$ (otherwise the moment does not exist).

[◀ Return](#)